

POKHARA ENGINEERING COLLEGE
Internal Assessment Examination

Level: Bachelor
Programme: Computer 7th
Course: Engineering Economics

Year : 2026
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Engineering Economics. Explain its principles. 2+5
b) Define variable cost, fixed cost, sunk cost and manufacturing cost with suitable examples. 8
2. a) Mrs. Sharma is planning for her retired life and has 10 more years of service. She would like to deposit 20% of her salary, which is Rs 4000 at the end of first year and thereafter she wishes to deposit the amount with an annual increase of Rs 1000 for the next 9 years with an interest of 10%. Find the total amount at the end of 10th year. 7
b) A machine costing of Rs 500000 having estimated salvage value of Rs 50000 and useful life of 5 years. Calculate every year depreciation amount and book value under diminishing balance method and straight-line depreciation method. 8
3. a) Evaluate the following project under IRR assumption and also show unrecovered investment balance in tabular form and diagram. 7
Initial investment Rs 800000, Annual Revenue Rs 220000
Annual cost Rs 20000 Salvage value Rs 20000
MARR 10% Useful life 6 years.
b) Find both types of BC ratio and evaluate the project. 8
Initial investment Rs 300000, Annual Revenue Rs 50000
Annual cost Rs 5000 Salvage value Rs 60000
MARR 9% Useful life 10 years.
4. a) Project A and project B are mutually exclusive project. Select the best project under repeatability assumption using PW formulation. 7

Items	Project A	Project B
Initial investment	Rs 200000	Rs 300000
Annual revenue	Rs 75000	Rs 85000
Annual cost	Rs 10000	Rs 12000
Useful life	6 years	8 years
Salvage value	15000	20000
MARR	10% for both projects	

- b) Calculate simple and discounted payback period of the following project.

Initial investment Rs 1000000, Annual Revenue Rs 300000
 Annual cost Rs 20000 Salvage value Rs 20000
 MARR 12% Useful life 10 years.

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5. a) Perform sensitivity analysis of the following project over the range of $\pm 25\%$ in i) initial investment ii) annual revenue iii) salvage value and iii) useful life.

Initial investment Rs 300000, Annual Revenue Rs 55000
 Salvage value Rs 60000 Useful life 10 years.
 MARR 9%

Also construct the sensitivity table and plot in the sensitivity chart and find out the most sensitive parameter.

- b) ABC Hydro Company has following capital structure

Common stock Rs 1000000
 Preferred stock Rs 800000
 Bond Rs 2000000
 Total capital Rs 4000000

common stocks were netted for Rs 985, expected dividend and growth rate were Rs 11 and 7 percent respectively.

Before tax cost of debt is 15% and tax rate is 25%

Cost of preferred stock is 12%

Calculate weighted average cost of capital of the company

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Or, Define capital budgeting with its features.

6. a. What do you understand by inflation? Explain different measures of inflation. 7
- b. Suppose a company has decided to purchase a new machine having initial cost of Rs 18000 and by expecting 8-year useful life and having operating cost of Rs 3000 in 1st year. For remaining year operating cost increases per year 15% over previous year operating cost and salvage value declines each year by 20% from the previous year salvage value. Find the economic service life of this new machine when MARR is 10%. 8

Or, The annual equivalent of the defender and challenger were of in the table below. What is the best replacement strategy? where MARR=12%. The planning horizon of the project is 8 years.

Year (n)	1	2	3	4	5	6
AEC def	5300	5250	5400	5750	6200	6550
AEC chall	7700	6150	5700	5600	5675	5800

7. Write short notes on any two: 2x5
- a) Vat
 - b) Law of demand
 - c) Causes of depreciation
 - d) Common Stock

The end